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Turning Code Into Compassion: A Visualize Aging Tool for Awareness and Reflection (AVATAR)



INTRODUCTION

Turning Code Into Compassion

The team experienced a gerontology (aging) suit that simulated vision impairment, muscle loss, and restricted mobility. Navigating everyday tasks highlighted how deeply functional decline impacts daily life. This inspired AVATAR: a visualization-driven platform designed to build empathy and foster self-reflection among young adults.



Fig. 1. Team members wearing an gerontology suit to experience aging

Visualize Your Future-Self

When trying to make healthier choices now, thinking about future consequences can seem too far off to have an impact. Research has found that personalized facial aging can increase both risk awareness and intentions to take protective action (Brinker et al., 2020). However, most past solutions only targeted one specific behavior and one future outcome.

Aging is MORE than just skin deep

Aging affects both the body's structure and its surface. From facial structure to skin texture, tone, elasticity, pigmentation, and muscle definition, there are average patterns that change how we all look as we get older. These changes also differ by individual, gender, ethnicity, and environment (Flament et al., 2020; Swift et al., 2021).

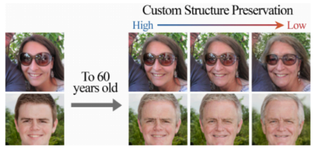


Fig. 2. Example from prior face-aging research showing identity-preserving age progression. Effective aging visualization must preserve recognizable facial identity while reflecting realistic age-related changes.

What AVATAR Adds

We combined principles of empathic modeling with biomedical aging research and identity-preserving AI. Instead of staring down one predetermined fate, users are shown a variety of potential futures they can relate to—making it easier for users to connect daily habits to the future they will one day face.

AIM

To develop AVATAR, a personalized aging simulator that enables young people to envision their future self in an authentic, deeply resonant way, fostering empathy and positive attitudes toward aging by helping them build a meaningful relationship with who they will become.

METHOD

Part 1. Biomedical Aging Knowledge Construction

To ensure AVATAR accurately reflects realistic aging mechanisms, we implemented a structured three-stage knowledge extraction pipeline:

Literature Review: Conducted a targeted synthesis of dermatological and anatomical studies focused on intrinsic and extrinsic aging

***Knowledge Extraction*:** Isolated specific physiological phenotypes linked to modifiable lifestyle habits and demographics

***Knowledge Standardization*:** Converted varied findings from different research studies into a single chart. This mapped specific lifestyle factors and ethnicities to exact facial regions and visual signs of aging

Part 2. Knowledge-Guided Prompt Generation

To translate biomedical knowledge into aging representation, we developed a four-step knowledge-guided generation pipeline:

***Knowledge Graph*:** Organized a standardized aging knowledge graph that models relationships between lifestyle factors, facial aging features and affected facial regions

Reasoning Engine: Propagated user lifestyle information through the knowledge graph to produce a structured aging representation

Prompt Construction: Ranked, filtered and merged the aging features into a concise natural language prompt while preserving identity

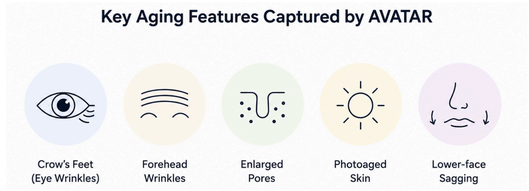
Image Generation: Applied the generated prompt to a pretrained image generation model to synthesize a personalized visualization

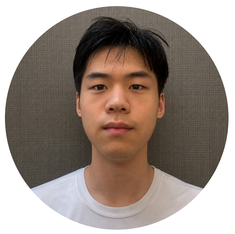
RESULTS

The input portrait (age 20) is compared with a baseline aging result and two AVATAR-generated results under different lifestyle profiles.

While the baseline reflects a typical 65-year-old appearance, AVATAR propagates lifestyle-related risks through the knowledge graph to produce personalized and progressively pronounced aging patterns.

Representative aging features inferred from the knowledge graph are then incorporated into the final prompt before image generation.





Input (age 20)

Fig. 3a



Baseline (age 65)

Fig. 3b



AVATAR: Light Unhealthy Profile

Fig. 3c



AVATAR: Heavy Unhealthy Profile

Fig. 3d

CONCLUSION

- **Goal Achieved:** Delivers personalized, identity-preserving visualizations grounded in biomedical research
- **Empathic Connection:** Replaces generic aging models with multi-variable lifestyle progression, helping young adults meaningfully visualize their future selves
- **Drives Health Literacy:** Connects daily lifestyle choices to long-term visual outcomes

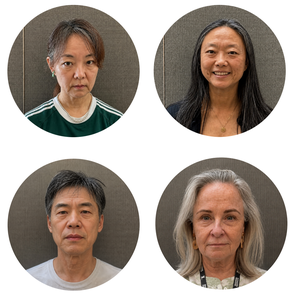


Fig. 4. Team members visualized at age 65

FUTURE WORK

As our empathic modeling tool is currently in its initial phase, future developments will focus on:

- **Algorithmic Refinement:** Redistributing propagation weight & updating knowledge engine
- **Graph Refinement:** Adding interaction & mechanism layer
- **Variable Expansion:** Integrating additional lifestyle metrices, such as sleeping habits and pollution
- Submitting paper to **ACM / IEEE 2027**
- Continue working through **SPIN 2026-2027**

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REFERENCES

AVATAR

BECKMAN INSTITUTE

(dis)ability DESIGN STUDIO

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